

A Comparative Study of Machine Learning, Deep Learning, and Foundation Models for Short-Term Electricity Load Forecasting

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Abstract

Short-term electricity load forecasting is a core component of smart-grid operation, predictive planning, and industrial energy management. This paper presents a reproducible benchmark comparing a seasonal baseline, XGBoost, three neural architectures, and two time-series foundation models. The primary evaluation uses the ECL electricity consumption dataset with aggregate load as the target. ETTh1 is used as a secondary industrial benchmark with HUFL, a load-related transformer variable, as the target; the oil-temperature variable OT is deliberately not treated as load. Experiments cover horizons of 24, 96, and 168 hours using a fixed 336-hour context. On ECL, TimesFM obtains the lowest point-wise MAE at H=24, while XGBoost is best at H=96 and H=168 with low single-window latency. Chronos-Bolt is consistently close to XGBoost without task-specific training. On ETTh1/HUFL, Chronos-Bolt is best at H=24, while DLinear is best at H=96 and H=168.

Keywords: short-term load forecasting; time series; deep learning; foundation models; XGBoost; smart grid

1 Introduction

Short-term load forecasting (STLF) supports dispatch planning, balancing, tariff-aware operation, maintenance planning, and anomaly-aware industrial energy management. In modern smart-grid and Industry 4.0 settings, forecasting models are expected not only to be accurate, but also to be reproducible, computationally affordable, and robust across operating regimes.

The methodological landscape is broad. Classical methods such as ARIMA and seasonal naive baselines remain common references; machine-learning models such as gradient boosting are attractive because they incorporate lag and calendar features with modest engineering cost; and recent deep-learning and foundation models promise broader transfer across time-series domains. This creates a practical question: when do expensive architectures or zero-shot foundation models actually improve over strong inexpensive baselines?

This paper reports a reproducible execution of the benchmark on two public datasets. The contribution is pragmatic: an end-to-end repository with data validation, leakage-aware scaling, metrics, timing, figures, paper generation, real results for seven models, Diebold–Mariano tests, and an accuracy–latency analysis.

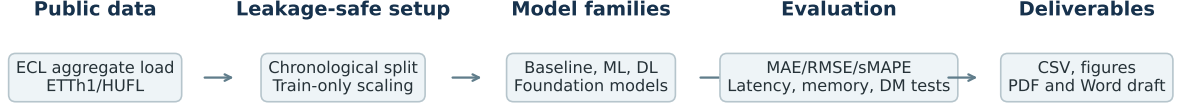


Figure 1: Experimental pipeline.

2 Related Work

Classical time-series forecasting methods, including ARIMA-family models, remain important because they establish interpretable and inexpensive references. Machine-learning approaches based on lagged targets, calendar variables, and gradient-boosted trees are widely used in load forecasting because they are fast to train and robust under tabular feature engineering. Deep-learning models such as LSTM, attention-based architectures, Informer, DLinear, PatchTST, iTransformer, and TimesNet have broadened the long-horizon forecasting toolbox.

Time-series foundation models such as TimesFM and Chronos motivate a new evaluation axis: no-task-specific-training inference versus supervised training on the target dataset. A fair comparison must state that public benchmarks may have appeared in pretraining corpora, so zero-shot performance on ECL or ETT should not be interpreted as guaranteed dataset-unseen generalization.

3 Methodology

Given a univariate context window $\mathbf{x}_{t-L+1:t}$, the task is to predict $\hat{\mathbf{y}}_{t+1:t+H}$ for horizons $H \in \{24, 96, 168\}$. The executed experiments use a fixed context length of $L = 336$ hours, which includes daily and weekly seasonal information and permits use of lag-168 features.

The SeasonalNaive baseline repeats the most recent daily seasonal pattern. XGBoost is trained as a direct multi-output regressor using lag-1, lag-24, lag-168, rolling means over 24 and 168 hours, and cyclic calendar features. DLinear, PatchTST, and iTransformer are trained through NeuralForecast in the same target-only setting, with a short validation-only learning-rate sweep over $\{10^{-3}, 5 \cdot 10^{-4}\}$ and early stopping. Chronos-Bolt-Small and TimesFM 2.5 are evaluated in zero-shot mode without task-specific training.

4 Experimental Setup

The primary dataset is ECL, processed as an aggregate hourly electricity load across 321 clients. The secondary dataset is ETTh1 with HUFL as the target. ETTh1 OT is oil temperature and is not used as electricity load. ECL is split chronologically into 70% train, 10% validation, and 20% test. ETTh1 uses the standard 12-month, 4-month, 4-month chronological split.

Table 1: Accuracy on ECL test split.

Model	H	MAE	RMSE	sMAPE (%)	wMAPE (%)
timesfm	24	259345.28	307450.43	31.27	29.60
chronos_bolt_small	24	259546.11	311466.37	31.32	29.63
xgboost	24	259549.38 $\pm$ 13.62	306563.19 $\pm$ 18.53	31.33 $\pm$ 0.00	29.63 $\pm$ 0.00
dlinear	24	278309.93 $\pm$ 78.45	327075.05 $\pm$ 121.45	34.58 $\pm$ 0.01	32.94 $\pm$ 0.01
patchtst	24	279295.35 $\pm$ 274.56	328419.63 $\pm$ 622.11	34.68 $\pm$ 0.04	33.05 $\pm$ 0.03
itransformer	24	283321.32 $\pm$ 278.79	335735.87 $\pm$ 639.05	35.11 $\pm$ 0.03	33.53 $\pm$ 0.03
seasonal_naive	24	338935.90	428088.82	39.98	38.69
xgboost	96	261792.58 $\pm$ 9.36	308906.88 $\pm$ 25.05	31.63 $\pm$ 0.00	29.94 $\pm$ 0.00
chronos_bolt_small	96	262601.94	314012.89	31.68	30.03
timesfm	96	262766.32	311382.39	31.65	30.05
dlinear	96	279227.54 $\pm$ 52.48	327146.57 $\pm$ 54.07	34.69 $\pm$ 0.01	33.07 $\pm$ 0.01
itransformer	96	279842.21 $\pm$ 148.37	327663.71 $\pm$ 205.90	34.75 $\pm$ 0.02	33.14 $\pm$ 0.02
patchtst	96	280109.77 $\pm$ 94.31	328333.77 $\pm$ 256.71	34.77 $\pm$ 0.02	33.17 $\pm$ 0.01
seasonal_naive	96	340658.82	428618.32	40.13	38.96
xgboost	168	264220.96 $\pm$ 24.77	311145.57 $\pm$ 42.08	31.96 $\pm$ 0.00	30.30 $\pm$ 0.00
chronos_bolt_small	168	264268.27	315996.95	31.91	30.30
timesfm	168	265044.60	313676.28	31.95	30.39
itransformer	168	279476.02 $\pm$ 103.81	326253.85 $\pm$ 265.79	34.75 $\pm$ 0.01	33.15 $\pm$ 0.01
dlinear	168	280272.56 $\pm$ 47.26	328134.31 $\pm$ 38.84	34.84 $\pm$ 0.01	33.25 $\pm$ 0.01
patchtst	168	280876.20 $\pm$ 109.22	328943.70 $\pm$ 263.11	34.88 $\pm$ 0.01	33.32 $\pm$ 0.01
seasonal_naive	168	343510.17	431537.72	40.48	39.39

Accuracy is measured with MAE, RMSE, sMAPE, and wMAPE. Efficiency is measured with training time, single-window inference latency, trainable parameters, and peak GPU memory. Diebold–Mariano tests are computed on per-window absolute errors for the top two models by MAE at seed 42 for each dataset and horizon.

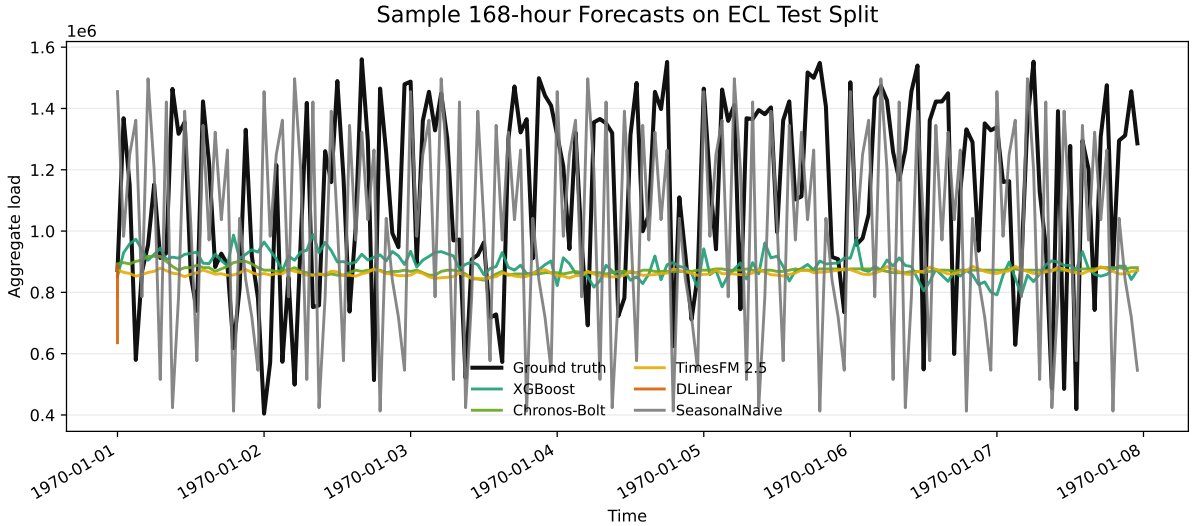


Figure 2: Seven-day ECL test segment.

Table 2: Accuracy on ETTH1 test split.

Model	H	MAE	RMSE	sMAPE (%)	wMAPE (%)
chronos_bolt_small	24	3.00	4.78	44.88	29.73
patchtst	24	3.04 ± 0.01	4.56 ± 0.01	47.47 ± 0.18	30.09 ± 0.11
dlinear	24	3.08 ± 0.00	4.63 ± 0.00	47.95 ± 0.10	30.53 ± 0.04
timesfm	24	3.16	4.88	48.33	31.25
itransformer	24	3.25 ± 0.02	4.77 ± 0.04	49.29 ± 0.09	32.17 ± 0.15
seasonal_naive	24	3.31	5.44	46.07	32.78
xgboost	24	3.89 ± 0.01	5.40 ± 0.01	57.47 ± 0.08	38.50 ± 0.06
dlinear	96	3.55 ± 0.00	5.26 ± 0.01	54.11 ± 0.07	35.08 ± 0.02
chronos_bolt_small	96	3.57	5.61	51.08	35.30
patchtst	96	3.62 ± 0.03	5.27 ± 0.03	54.87 ± 0.36	35.79 ± 0.28
timesfm	96	3.67	5.61	54.27	36.24
itransformer	96	3.72 ± 0.01	5.43 ± 0.02	55.56 ± 0.24	36.81 ± 0.12
seasonal_naive	96	3.92	6.41	51.70	38.80
xgboost	96	4.21 ± 0.01	5.86 ± 0.01	61.21 ± 0.04	41.63 ± 0.05
dlinear	168	3.70 ± 0.00	5.43 ± 0.01	56.49 ± 0.08	36.54 ± 0.02
chronos_bolt_small	168	3.77	5.85	53.36	37.20
timesfm	168	3.79	5.75	55.77	37.38
itransformer	168	3.91 ± 0.02	5.62 ± 0.03	57.87 ± 0.23	38.62 ± 0.23
patchtst	168	3.95 ± 0.07	5.62 ± 0.08	58.51 ± 0.74	38.96 ± 0.74
seasonal_naive	168	4.18	6.70	54.20	41.26
xgboost	168	4.27 ± 0.01	5.98 ± 0.01	61.54 ± 0.04	42.15 ± 0.05

5 Results

On ECL, the strongest group consists of XGBoost and the two foundation models. TimesFM obtains the lowest point-wise MAE at H=24, while XGBoost is best at H=96 and H=168. Chronos-Bolt is within approximately 0.3% of the best MAE at all three horizons. DLinear, PatchTST, and iTransformer improve substantially over SeasonalNaive, but they do not match the tree model or foundation models under the current training budget.

On ETTh1/HUFL, Chronos-Bolt is best at H=24, PatchTST is the strongest trained model at H=24, and DLinear is best at H=96 and H=168. TimesFM is competitive but not the best on this benchmark.

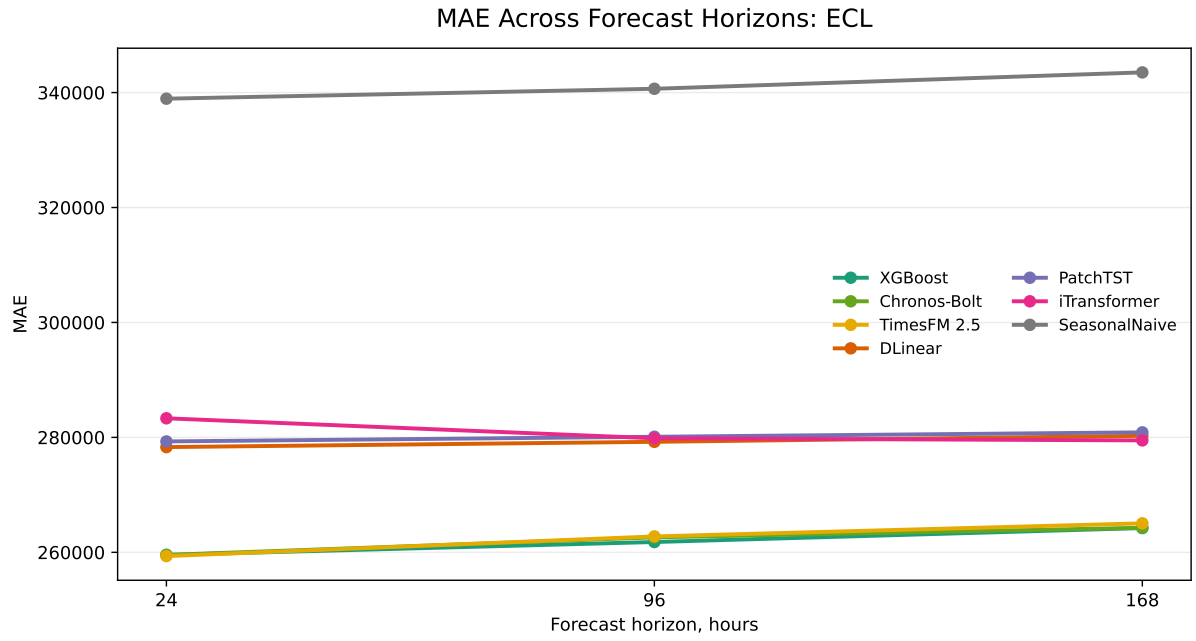


Figure 3: MAE across forecast horizons on ECL.

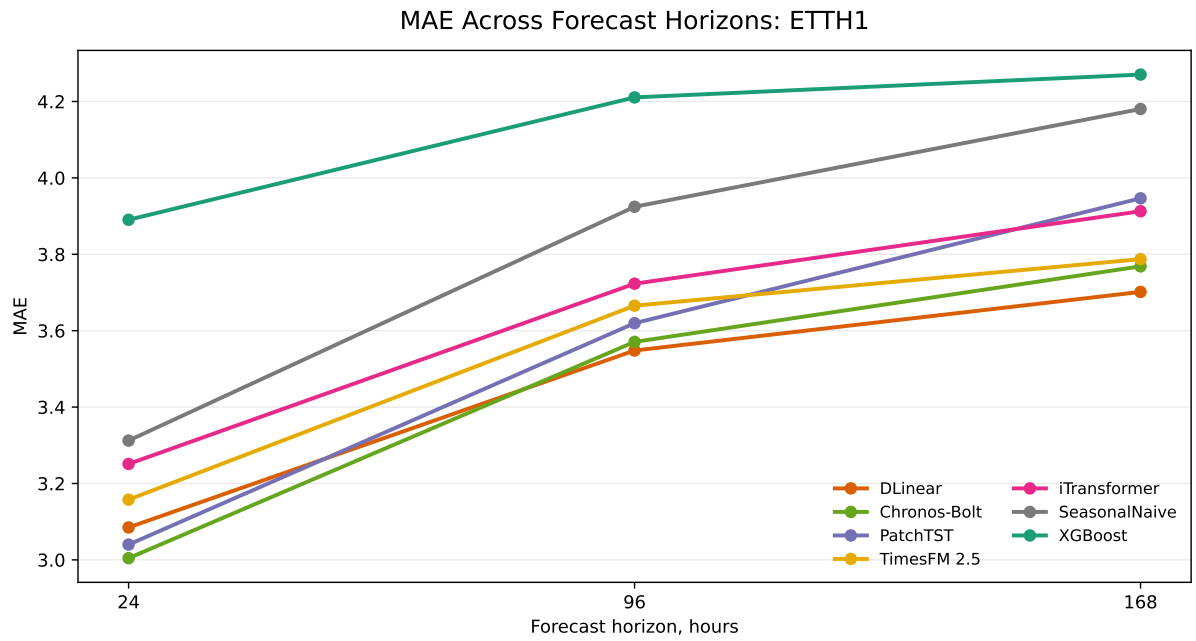


Figure 4: MAE across forecast horizons on ETTh1/HUFL.

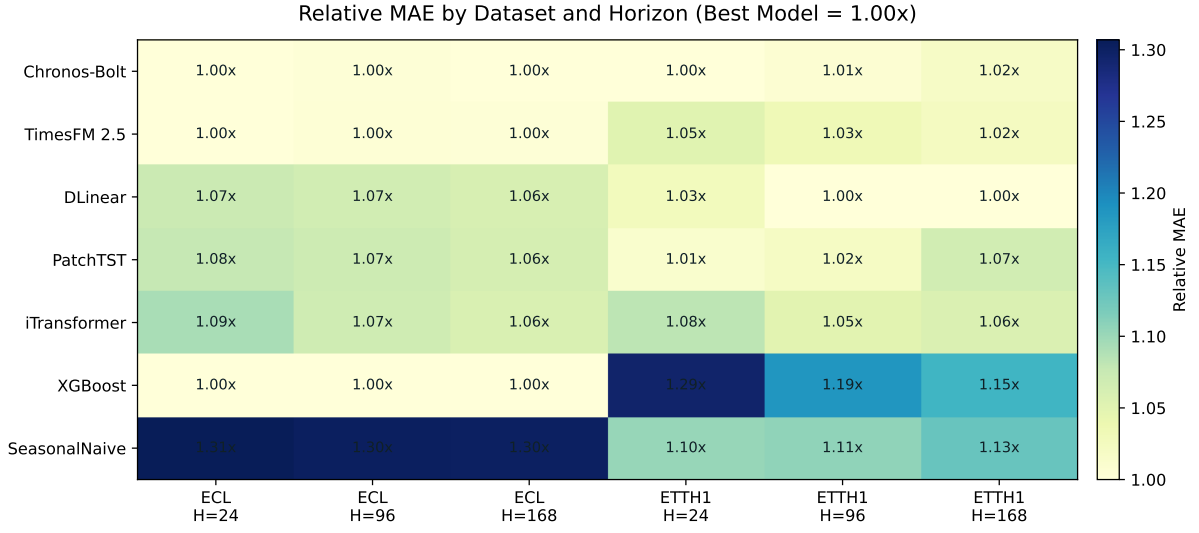


Figure 5: Relative MAE heatmap. Values are normalized within each dataset and horizon.

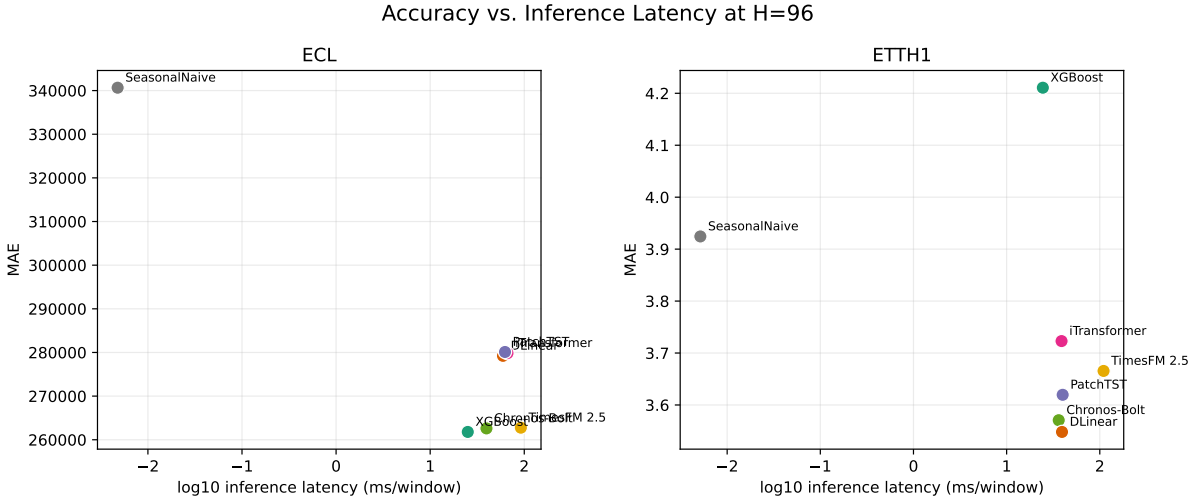


Figure 6: Accuracy–latency view at H=96.

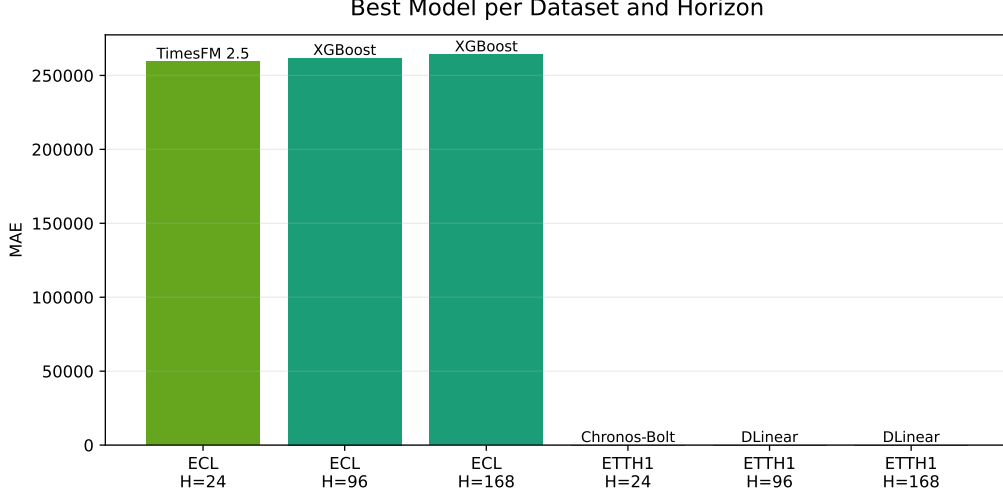


Figure 7: Best model per dataset and horizon.

6 Discussion

The results show that model choice is dataset-dependent. On aggregate ECL load, engineered lag features with XGBoost remain an extremely strong and inexpensive option. Foundation models are attractive when avoiding task-specific training is valuable: Chronos-Bolt is close to XGBoost on ECL and best at the shortest ETTh1/HUFL horizon. DLinear is the strongest trained neural model in this benchmark and dominates the longer ETTh1/HUFL horizons.

From a deployment perspective, the models occupy different operating regimes. XGBoost is the most attractive choice when a conventional supervised training pipeline is acceptable: it is accurate on ECL, has low single-window latency, and does not require GPU inference. Chronos-Bolt is useful when task-specific training is undesirable, for example in cold-start industrial monitoring, rapid prototyping, or sites where historical data cannot be pooled for model training. TimesFM shows strong point accuracy at the shortest ECL horizon, but its latency is substantially higher in this implementation. DLinear is a good neural reference model because it is simple, compact, and strong on the longer ETTh1/HUFL horizons.

The transformer results should be read carefully. PatchTST and iTransformer are credible architectures, but this benchmark uses a compact target-only configuration and a limited training budget. Their underperformance here does not imply that transformers are generally unsuitable for load forecasting; rather, it shows that they do not automatically beat simpler methods under constrained, reproducible settings.

7 Limitations

The main limitations are public-benchmark pretraining-contamination risk for foundation models, limited hyperparameter search, target-only modeling, and the absence of private industrial deployment data. A rolling AutoARIMA baseline was attempted, but full per-origin refitting was prohibitively expensive in this protocol; the partial run is retained in the repository logs but excluded from the main comparison tables. Chronos-Bolt also emits a warning that predic-

tion lengths above 64 are outside its recommended range; therefore, $H=96$ and $H=168$ Chronos results should be interpreted as practical stress tests rather than optimal use of that model.

The foundation-model results should not be overinterpreted as guaranteed dataset-unseen zero-shot generalization. ECL and ETT are public benchmarks and may overlap with pre-training or evaluation corpora used during model development. The correct claim is narrower: Chronos-Bolt and TimesFM were run without task-specific training in this repository.

8 Conclusion

We built and executed a reproducible STLTF benchmark pipeline on ECL and ETTh1/HUFL. XGBoost is the most practical choice on ECL when training data are available, TimesFM slightly leads $H=24$ point-wise MAE, and Chronos-Bolt provides competitive zero-shot accuracy. On ETTh1/HUFL, Chronos-Bolt wins $H=24$, PatchTST is the best trained model at $H=24$, and DLinear wins $H=96$ and $H=168$. These findings support a pragmatic recommendation: compare against strong simple baselines first, use DLinear and PatchTST as neural references, and treat foundation models as valuable cold-start tools rather than universally dominant replacements for supervised forecasting.

Table 3: Computational efficiency.

Model	Data/H	Train s	Infer ms	Params	GPU MB
chronos_bolt_small	ecl/H24	0.000	15.6205	47718016	123.3
dlinear	ecl/H24	11.795	55.8597	16176	26.3
itransformer	ecl/H24	23.725	55.3372	311448	164.9
patchtst	ecl/H24	43.077	55.7089	168280	608.6
seasonal_naive	ecl/H24	0.000	0.0041	0	0.0
timesfm	ecl/H24	0.000	102.2329	200000000	895.6
xgboost	ecl/H24	2.892	7.5029	57388	0.0
chronos_bolt_small	ecl/H96	0.000	39.6397	47718016	345.8
dlinear	ecl/H96	18.843	59.4035	64704	29.8
itransformer	ecl/H96	31.380	66.2948	320736	165.0
patchtst	ecl/H96	49.984	62.4020	361888	612.4
seasonal_naive	ecl/H96	0.000	0.0048	0	0.0
timesfm	ecl/H96	0.000	91.9861	200000000	895.6
xgboost	ecl/H96	10.485	25.0947	229423	0.0
chronos_bolt_small	ecl/H168	0.000	45.2988	47718016	381.4
dlinear	ecl/H168	25.996	64.1642	113232	32.7
itransformer	ecl/H168	38.540	69.7238	330024	165.2
patchtst	ecl/H168	56.942	75.2900	555496	617.3
seasonal_naive	ecl/H168	0.000	0.0058	0	0.0
timesfm	ecl/H168	0.000	200.3501	200000000	896.4
xgboost	ecl/H168	18.179	42.7073	401454	0.0
chronos_bolt_small	etth1/H24	0.000	13.4838	47718016	123.3
dlinear	etth1/H24	10.701	39.9808	16176	26.2
itransformer	etth1/H24	20.171	39.1519	311448	164.8
patchtst	etth1/H24	39.820	39.1007	168280	608.3
seasonal_naive	etth1/H24	0.000	0.0037	0	0.0
timesfm	etth1/H24	0.000	98.1902	200000000	895.6
xgboost	etth1/H24	2.340	6.1449	57417	0.0
chronos_bolt_small	etth1/H96	0.000	36.2363	47718016	345.8
dlinear	etth1/H96	9.899	39.2943	64704	29.8
itransformer	etth1/H96	20.615	38.8574	320736	165.0
patchtst	etth1/H96	38.598	39.9531	361888	612.6
seasonal_naive	etth1/H96	0.000	0.0052	0	0.0
timesfm	etth1/H96	0.000	109.7588	200000000	895.6
xgboost	etth1/H96	8.638	24.4694	229758	0.0
chronos_bolt_small	etth1/H168	0.000	64.7316	47718016	381.4
dlinear	etth1/H168	10.095	40.7118	113232	32.6
itransformer	etth1/H168	21.127	44.3775	330024	165.1
patchtst	etth1/H168	39.072	42.3925	555496	617.0
seasonal_naive	etth1/H168	0.000	0.0053	0	0.0
timesfm	etth1/H168	0.000	208.5066	200000000	896.4
xgboost	etth1/H168	14.775	42.6873	401899	0.0

Table 4: Diebold–Mariano tests for top-2 models by MAE at seed 42.

Data/H	Comparison	Statistic	p-value	Sig.
ecl/H24	timesfm vs xgboost	13.45	¡0.001	True
ecl/H96	xgboost vs chronos_bolt_small	-38.15	¡0.001	True
ecl/H168	xgboost vs chronos_bolt_small	-2.88	0.004	True
etth1/H24	chronos_bolt_small vs patchtst	-3.39	¡0.001	True
etth1/H96	dlinear vs chronos_bolt_small	-2.48	0.013	True
etth1/H168	dlinear vs chronos_bolt_small	-8.81	¡0.001	True